Abstract Algebra | Quiz 8 | October 28, 2026

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1. Suppose $\varphi : \mathbb{Z} \to \mathbb{Z}_5 \times \mathbb{Z}_6$ is a homomorphism, and $\varphi(1) = (3, 2)$.

   (a) $\varphi(3) = \varphi(3 \cdot 1) = 3 \cdot \varphi(1) = 3 \cdot (3, 2) = (9, 6) = \boxed{(4, 0)}$

   (b) $\ker(\varphi) = \boxed{\langle 15 \rangle \subseteq \mathbb{Z}}$

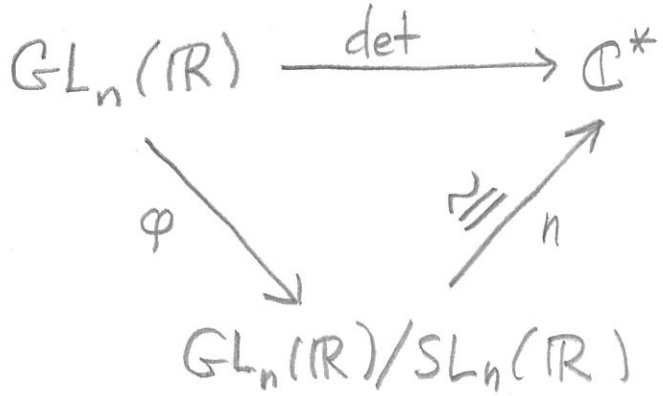
   Because $\varphi(n) = \varphi(n \cdot 1) = n \cdot \varphi(1) = n(3, 2)$ $= (3n, 2n) \in \mathbb{Z}_5 \times \mathbb{Z}_6$, it follows that $\varphi(n) = (3n, 2n)$ equals $(0, 0)$ only if n is both a multiple of 5 and a multiple of 3, that is, if n is a multiple of 15, which means $n \in \langle 15 \rangle \subseteq \mathbb{Z}$. Hence $\ker(\varphi) = \langle 15 \rangle$.

2. Recall that the determinant function $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$ is a surjective homomorphism.

   (a) $\ker(\varphi) = \{ A \in GL_n(\mathbb{R}) : \det(A) = 1 \}$

   $= \boxed{SL_n(\mathbb{R})}$

   (b) According to the First Isomorphism Theorem, $GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \boxed{\mathbb{R}^*}$.



3. Prove that a homomorphism $\varphi : G \to H$ is injective if and only if $\ker(\varphi) = \{e\}$.

Proof

($\Rightarrow$) Suppose φ is injective. We know that $\varphi(e_G) = e_H$. Now, if $x \in G$ and $\varphi(x) = e_H$, then

$$\varphi(x) = \varphi(e_G),$$

so $x = e_G$ because φ is injective. In other words, the only element of G that φ sends to e_H is e_G. This means $\ker(\varphi) = \{e_G\} = \{e\}$.

($\Leftarrow$) Conversely suppose $\ker(\varphi) = \{e\}$. We must show φ is injective. Suppose $\varphi(x) = \varphi(y)$ then

$$\varphi(x)(\varphi(y))^{-1} = e_H$$
$$\varphi(x)\varphi(y^{-1}) = e_H$$
$$\varphi(xy^{-1}) = e_H.$$

But because $\ker(\varphi) = \{e_G\}$ we must have $xy^{-1} = e_G$, and so $x = y$, meaning φ is injective